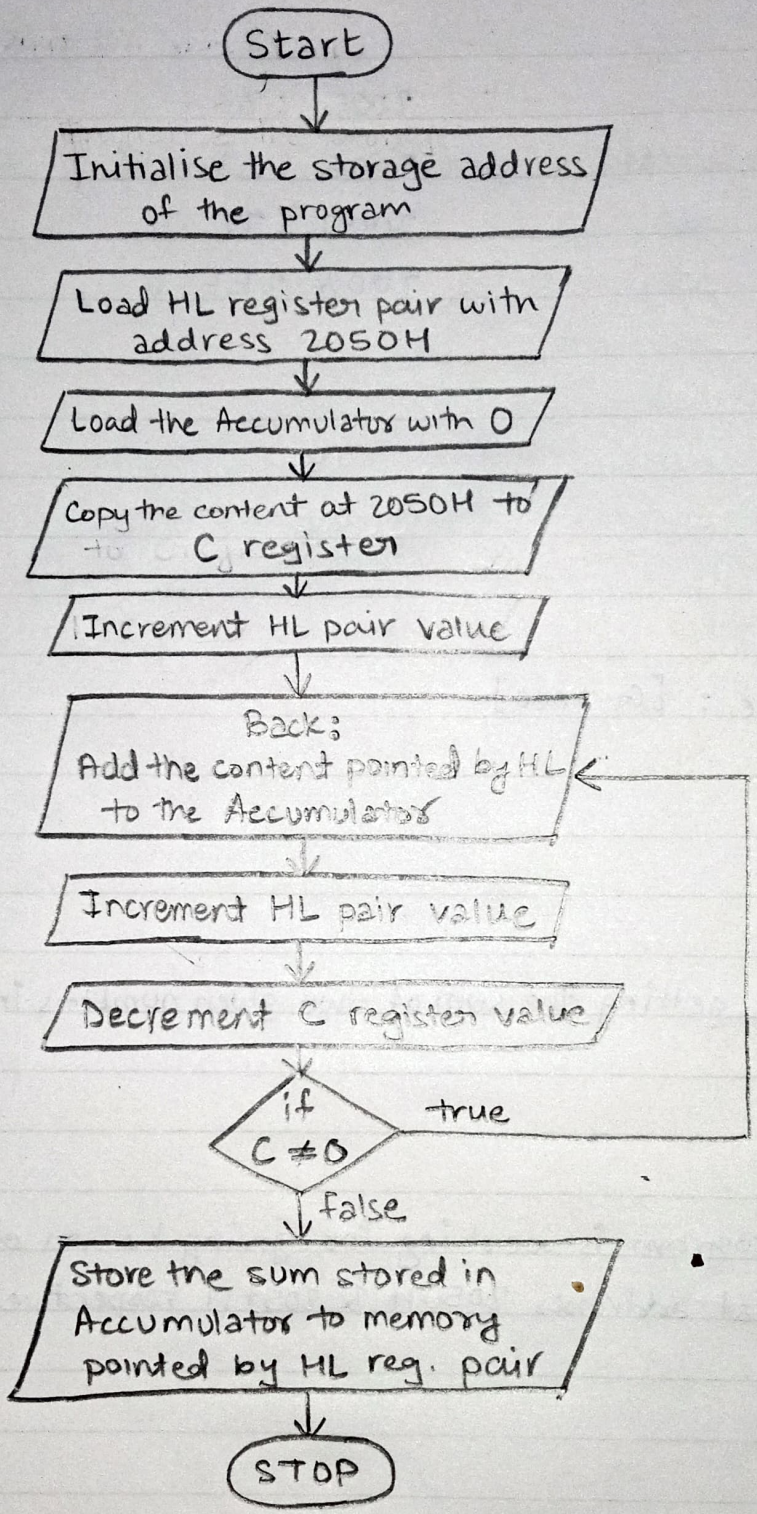
• **Flow chart :**


Start
Initialise the storage address of the program
Load HL register pair with address 2050H
Load the Accumulator with 0
Copy the content at 2050H to C register
Increment HL pair value
Back;
Add the content pointed by HL to the Accumulator
Increment HL pair value
Decrement C register value
if C ≠ 0
true
false
Store the sum stored in Accumulator to memory pointed by HL reg. pair
STOP

Date 26/07/22

Expt. No. 2 Page No.

- **Aim:** Write a program in assembly language of 8085 to find 8-bit sum of N-8 bit numbers stored at consecutive memory location.

- **Apparatus:** 8085 training kit, keyboard

- **Algorithm:** (i) Start
  (ii) Initialise the storage address of the pragram.
  (iii) Load the HL register pair with address 2050H.
  (iv) Copy the value at 2050H to C register, it will be used as counter.
  (v) Increment HL register pair to next address
  (vi) Load the Accumulator with zero value
  (vii) Start of the Loop, add value pointed by HL register to Accumulator.
  (viii) Increment HL register pair to next address
  (ix) Decrement C register counter by 1
  (x) If value in C register doesn't become zero, repeat the loop
  (xi) else, store the sum from Accumulator to memory address pointed by HL regi pair.
  (x) Stop

- **Registers Used:** to reference memory address → HL register pair
  to store the sum → Accumulator
  to control loop count → C register.

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- **Observation Table:**

Memory add.	Before	After	Decimal eq.
2050	04	04	–
2051	01	01	1
2052	02	02	2
2053	03	03	3
2054	04	04	+4
2055	00	0A	10

26/7/22

Date _____

Expt. No. _____ Page No. _____

- **Code:**

	Mnemonic Code	Hex-code
	LXI H, 2050H	2000 : 21
		2001 : 50 , 2002 : 20
	MOV C, M	2003 : 4E
	INX H	2004 : 23
	MVI A, 00H	2005 : 3E , 2006 : 00
Back :	ADD M	2007 : 86
	INX H	2008 : 23
	DCR C	2009 : 0D
	JNZ Back	200A : C2
		200B : 07 , 200C : 20
	MOV M, A	200D : 77
	RST	200E : EF

- **Observation :** ~~This program adds one 8 bit numbers and store the sum at the next address and stores the carry to the next address~~

  Input data : 4, {01, 02, 03, 04}

  Output data : 0A [∵ 01+02+03+04 = 0A in hexa]

- **Result :** This program is working for finding 8-bit sum of N 8-bits numbers stored at consecutive memory locations, and stores the sum at next address.

Teacher's Signature : _____